Do methods matter in the meta-analysis of partial correlation coefficients?

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Abstract

Recent studies have demonstrated that conventional meta-analyses of partial correlation coefficients (PCC) are biased. Several adjustments have been shown in simulations to reduce these small-sample biases to negligibility. While many meta-analyses of partial correlation coefficients are conducted each year across several disciplines, the practical importance of these issues remains unknown. To address this question and to offer advice for applications, we survey 172 economic meta-analyses of PCCs. We find that small-sample biases are negligible in practice. However, some publication selection biases remain. Although random effects using Fisher's z transformations have often been recommended, they reduce neither small-sample nor publication selection biases relative to conventional random effects. The family of unrestricted weighted least squares (UWLS) and the Hunter-Schmidt (HS) estimators produce smaller, arguably less biased, estimates of the mean PCC in these applications than either random effects with or without Fisher's z transformations. These findings offer practical guidance for any discipline that meta-analyzes partial correlations.

Keywords: partial correlation coefficients, meta-analysis, small sample, bias, publication bias

Paper page: https://meta-analysis.cz/pcc_survey
Data and code: https://github.com/PetrCala/pcc-survey

Highlights

What is already known?

- Conventional meta-analyses of partial correlation coefficients (PCCs) are biased, although the biases are relatively small in most cases.
- Several adjustments have been proposed to reduce these small-sample biases to negligibility.
- Many meta-analyses of PCCs are conducted each year.

What is new?

- We survey the performance of alternative methods across 172 meta-analyses of PCCs.
- In practice, small-sample biases are trivial; however, likely publication selection bias remains.
- Although Fisher's *z* version of random effects is often recommended, it does not reduce estimated mean effects relative to conventional random-effects models.
- Both unrestricted weighted least squares (UWLS) and Hunter and Schmidt's approaches to the meta-analysis of PCCs produce more conservative estimates that are thereby consistent with being less affected by widely known upward biases than random effects in applications.

Potential impact for *Research Synthesis Methods* readers outside the authors' field

- Although our data come from economics, the key findings are likely to apply to other disciplines wherever PCCs are meta-analyzed with similarly heterogeneous sample sizes, including health sciences, education, and psychology.

1. INTRODUCTION

Meta-analysis is used across the disciplines to integrate, summarize, and understand different research findings for a given area of research, hypothesis, or treatment. Meta-analyses are statistical analyses of statistical findings that are regarded to be at the top of the "pyramid of evidence".[1,2] However, they require a fully comparable metric (an 'effect size') that quantifies the sometimes very different research findings across numerous studies of the same phenomenon.[3-5] In economics and other social sciences, it is common to find that different studies of the same conceptual effect employ different outcome measures, treatment metrics, dependent and/or independent variable measures of the phenomenon in question. To circumvent this comparability issue, partial correlation coefficients (PCCs) are frequently used in the meta-analysis of economics research (e.g.,[6-8]). PCCs are also increasingly used in environmental science, management, and public health, making the choice of meta-analysis methods relevant well beyond economics. As a unitless measure of statistical effect (a correlation), PCCs can be sensibly compared across different measures of the dependent variable (for example, per capita GDP, GDP, and GDP growth) and different metrics of the independent variable (e.g., broadband access, cell phone contracts, or IT expenditures when technology-growth is being studied).

Recent studies cast doubt on the proper use of partial correlations in meta-analyses. Van Aert and Goos[9] claimed that using the 'wrong' formula for PCC's variance, $\left(1-r_p^2\right)/df$, biases meta-analysis estimates of the mean effect. While Stanley and Doucouliagos[10] showed that conventional meta-analyses of PCCs (e.g. fixed and random effects) are biased, using this 'wrong' formula systematically and predictably reduces these biases. Usefully for practice, these biases tend to be small and become negligible with large primary study sample sizes. Stanley, Doucouliagos, and Havranek[11] find that both well-known and novel approaches to correcting publication selection bias (PSB) can reduce these small-sample biases to scientific negligibility.

These small-sample biases are caused by a mechanical correlation of PCC estimates with their variances, which, in turn, affects any inverse-variance weighted average. Traditionally, the solution to a correlation between effect sizes and their standard errors (SE) is to employ the Fisher's z transformation to random effects. Although simulations confirm that Fisher's z transformation can do much to reduce this bias, Stanley, Doucouliagos and Havranek[11] find that a

simple adjustment to degrees of freedom allows the unrestricted weighted least squares (UWLS) to reduce these biases further. Stanley, Doucouliagos and Havranek[12] have also shown that the well-known but infrequently employed meta-analysis method developed by Hunter and Schmidt[13] also works well to practically eliminate these same small-sample biases seen in the meta-analysis of bivariate correlations. Although these biases are well-established in theory and simulations, will they matter in practice? This is the central question addressed by this paper.

Empirical confirmation is important because no simulation can capture the full complexity of actual meta-analytic data, and simulation-based recommendations that fail in practice could mislead hundreds of applied meta-analyses each year. In particular, the typical area of research surveyed here uses a wide dispersion of sample sizes. The average of median sample sizes is 959 and the average interquartile range is 3,473, representing both large average sample sizes and wide dispersions among the sample sizes within the survey's typical PCC meta-analysis. This contrasts sharply with previous simulations where the sample size was held constant for each study in the meta-analysis, and yet the identified small-sample biases became trivial at sample sizes of 400 or larger.[10] In addition to the wide distribution of sample sizes, meta-analyses of PCCs typically have high heterogeneity and are often afflicted with notable PSB. Even though heterogeneity and PSB have been simulated in Stanley, Doucouliagos and Havranek,[11,12] the complexity and nuance of typical PCC meta-analyses make it essential to assess these alternative methods in practice.

From a large set of meta-analyses of economics research, we have identified 172 meta-analyses that were conducted using PCCs as the effect sizes.[14,15] Below, we investigate this large set of empirical economic effects (63,730) to see whether these alternative meta-analysis methods cause notable differences to the interpretation of the findings for these areas of research. Because hundreds of meta-analyses are conducted each year using PCCs, guidance about which methods are best is needed, yet no study has systematically compared these estimators across a large body of empirical meta-analyses.

The purpose of this paper is to investigate whether alternative meta-analysis methods matter, practically or scientifically, in the synthesis of partial correlations. Typically, economics meta-analyses involve hundreds of estimates, many of which come from studies that used hundreds, thousands, or even millions of observations. With large sample sizes, these small-sample biases are expected to become scientifically negligible. However, with both small- and

large-sample studies mixed together, the relative advantages or biases of alternative meta-analysis methods are unknown. We fill this gap in our understanding by investigating 172 economic meta-analyses containing 63,730 estimates and their standard errors (SE) to see if the use of alternative meta-analysis methods has any notable effects on the summary estimates of PCCs. In the process, we find that the typical effect in this literature is very small. The median PCC is approximately 0.046, which in many contexts would fall below the smallest effect size of interest.

2. META-ANALYSIS OF PARTIAL CORRELATIONS

Partial correlations are often employed to solve a common problem found in social science research. Different studies of the same economic phenomenon will frequently quantify the dependent variable and/or the independent variable using fundamentally different units of measurement. For example, per capita GDP, productivity increases, and GDP growth rates are all used by empirical studies when studying the effect that the further adoption of information and communication technology (ICT) has on a country's 'economic growth'.[7] Substantively different scales, metrics, or measures cannot be statistically or meaningfully combined. However, this "apples and oranges" dilemma conflicts with systematic reviews' imperative to include all relevant research findings on a given issue or phenomenon.[4,5,16,17]

Partial correlations offer a straightforward escape from this dilemma. Any estimated regression coefficient can be converted to a partial correlation coefficient, r_p, by:

$$r_p = {}^{t}\Big/_{\sqrt{t^2+df}}, \tag{1}$$

Gustafson.[18] t is the conventional t-test for the statistical significance of the effect in question (e.g., ICT growth) in the explanation of the dependent variable (e.g., GDP growth), and df is the degrees of freedom available to the estimated multiple regression. r_p measures the statistical strength and direction of a linear relationship between the dependent phenomenon (e.g., GDP growth) and the variable of interest (e.g., ICT), after factoring out the effects of other independent variables (e.g. trade and education). r_p^2 is the proportion of the variation in the dependent variable

attributable to variation in the target independent variable after accounting for the effects of all other independent variables.

Eq. (1) provides comparable effect sizes as long as the different dependent variable metrics (or independent variable metrics) are all considered to measure the same general economic phenomenon (e.g., economic growth). However, to conduct a meta-analysis, we still need the standard errors (SE) of these PCCs. Stanley and Doucouliagos[5] advocated the use of the same SE that is used to test whether the population partial correlation is zero (H_0: $r = 0$) because it is this test of statistical significance that will potentially be gamed for those who seek to report statistically significant findings (i.e., publication selection bias or PSB):

$$S_1 = \sqrt{\left(1-r_p^2\right)/df} \tag{2}$$

Stanley and Doucouliagos.[5(p25)]

Recently, Van Aert and Goos[9] criticized the recommendation of using Eq. (2) to calculate PCC's SE as the asymptotic variance of PCC's sampling distribution is widely known to be the square of:

$$S_2 = \sqrt{\left(1-r_p^2\right)^2/df} \tag{3}$$

Olkin and Siotani.[19] Thus, S_2 is the 'correct' SE if we confine our interest to the sampling variation of a *single* PCC estimate. However, the conventional 'correct' SE formula, by more faithfully capturing sampling variation, ironically induces larger inverse-variance weighting distortions and resulting meta-analysis biases. Thus, the 'correct' SE formula for an individual PCC estimate is the 'wrong' formula to use in meta-analysis. Furthermore, because $S_2 < S_1$ for all $|r_p| \neq \{0 \text{ or } 1\}$, S_2 can make some studies with statistically insignificant effects appear statistically significant. This is especially a problem in economics where there is often a jump in the frequencies of reported statistical tests just above the .05 threshold.[20,21]

Nonetheless, Van Aert and Goos[9] claim that using any formula other than Eq. (3) for PCC's SE "biases the results of meta-analysis." Stanley and Doucouliagos[10] show, to the contrary, that using S_2 in conventional meta-analysis (fixed effect, random effects, and unrestricted weighted least squares, UWLS) doubles the bias relative to S_1. Thus, for use in meta-analysis, $S_2$is the 'wrong' estimate of the standard error as S_2 consistently produces larger biases. However, these biases are small-sample biases that exaggerate estimated mean effect sizes but decrease with larger sample sizes. Below, we investigate whether these different standard error/variance formulas make a notable difference in *practice*.

Conventional meta-analysis estimates the overall mean effect for a given area of research by inverse variance weighted averages, called fixed and random effects. The fixed-effect estimate (FE) of *r* is a weighted average of all the PCCs in an area of research with weights, $1/SE_i^2$, where SE_ican represent either S_1or S_2. Fixed effects assume that there is a single common effect with no variation in 'true' effects. In economics, research results are known to be conditional upon different economic systems, institutions, laws, histories, customs, etc., creating systematic heterogeneity that is almost always identified by multiple meta-regression. Thus, FE should be rarely and cautiously used to summarize economics research.[i] Furthermore, observed heterogeneity is typically several times larger than what can be accounted for by SEs alone.[23]

Random effects (RE) explicitly accommodate heterogeneity using weights, $(1/(SE_i^2+\hat{\tau}^2))$, that include an estimate of the heterogeneity variance, $\hat{\tau}^2$. Because $\hat{\tau}^2$ is constant across all of the estimates within a meta-analysis, random-effects weights are moderated from the upweighting of the more powerful and more precise, $1/SE_i$, estimates. However, many studies have shown that RE is more biased in the presence of PSB than FE and the unrestricted weighted least squares (discussed below), and there is much evidence that publication bias is often substantial in economics.[5,14,15,20,23-32]

Like FE, the unrestricted weighted least squares (UWLS) uses inverse variance weights ($1/SE_i^2$). Unlike FE, UWLS does not assume a common effect and estimates a proportional variance, γ, which mechanically and automatically adjusts for any excess heterogeneity when present. UWLS is estimated by α_1 in the simple meta-regression model:

$$t_i = \frac{r_{p_i}}{SE_i} = \alpha_1\left(\frac{1}{SE_i}\right) + u_i \qquad i = 1, 2, \ldots, k \qquad (4)$$

Where *k* is the number of PCCs combined into the meta-analysis. Any standard regression software automatically reports UWLS, $\hat{\alpha}_1$, its standard error, *t*-test, and confidence intervals when: *t*-values, r_{p_i}/SE_i, are specified as the dependent variable, precision, $1/SE_i$, is the only independent variable, and there is no intercept. Although UWLS will have the same point estimate as FE, the difference is that UWLS does *not* assume a common effect and will automatically accommodate observed between-study multiplicative heterogeneity, when present, in its estimated variance, $\hat{\gamma}/\Sigma\,1/SE_i^2$; where $\hat{\gamma}$ is the MSE of meta-regression Eq. 4. UWLS is a variety of random effects where the estimates' variances are allowed to be proportional to SE_i^2.

Some have associated UWLS with the Knapp-Hartung (KH) method of adjusting RE's standard error and confidence interval to moderate $\hat{\tau}^2$'s well-known bias.[33,34] van Aert and Jackson[35] justify KH as a weighted-least-squares estimator in which the "error variances are known up to a constant of proportionality" (p.521), conditionally, after estimating RE's additive heterogeneity variance. UWLS and KH, therefore, both rescale a known or estimated variance by an estimated multiplicative constant (sometimes called an overdispersion parameter), but they begin with different variances. UWLS assumes $\mathrm{Var}(\varepsilon_i) = \gamma \cdot SE_i^2$, weights by $1/SE_i^2$, and SE_i^2 are data provided by the primary studies along with their associated effect sizes. KH starts with the same primary study data but must first estimate the mean and τ^2 assuming additive random-effects variances. KH further assumes a multiplicative overdispersion factor that rescales RE's estimated variance by the weighted mean squared residual that uses RE's weights. Although KH and UWLS share the general WLS approach to estimating excess variation, they otherwise differ in their variance specifications and weights. UWLS and KH have different estimates of the mean effect with different uncertainties reflected in their different SEs, CIs, and test statistics, with the lone exception that the mean estimates are all exactly equal for FE, RE, UWLS, and KH when there is no heterogeneity (i.e., $\tau^2 = 0$). KH's overdispersion estimate will also generally differ from UWLS' γ.

In contrast to KH, UWLS is a fully formed classical regression model when variances are assumed to be proportional to SE_i^2. As such and without requiring normality, UWLS's estimates

are BLUE (smallest variance, 'best,' linear, and unbiased estimator) by the Gauss-Markov theorem.[36-38] With normal errors, UWLS's mean and variance estimators are also independent. These desirable properties are, of course, conditional upon the assumed model(s) being correct and do not establish that UWLS is superior to RE or KH, which maintain different model and variance structures. However, KH and RE cannot be proved to be BLUE or to have independent mean and variance estimates because the heterogeneity variance must be estimated in practice.[38] The desirable statistical properties of all of these estimators no longer hold under publication selection, which makes effects dependent upon their standard errors.

All three of these meta-analysis estimators of mean effect (RE, FE and UWLS) have small-sample biases, regardless of which formula for SE is used.[10,11] The reason for these biases is that the weights themselves depend on the estimated value of PCC, r_p. As discussed above, all conventional meta-analyses employ inverse-variance weights. Because PCC variances are proportional to $\left(1 - r_p^2\right)$, or its square, recall Eqs. (2) and (3), positive sampling errors will receive larger weight, thereby biasing the meta-analysis estimate upwards.[10,11]

An obvious remaining question is: how can these biases be eliminated or at least reduced to scientific negligibility? Stanley, Doucouliagos and Havranek[11,12] offer several approaches to reduce these biases to scientific triviality. However, it should be stressed that these are *small-sample* biases; thus, they are virtually zero (i.e., < .01) without any adjustment whenever $n_i \geq 200$. Because many social science regressions are estimated from more than 200 observations, often very much more, it is an empirical question whether these biases should concern meta-analysts in these fields at all. Answering this question is a central purpose of this study. To do so, we must also consider alternative adjustments that reduce these small sample biases to scientific triviality. Lastly, it is important to note that these small-sample and publication selection biases both tend to increase the magnitude of the reported PCCs, trends that have been verified in simulations and are reflected in our survey results below. Thus, smaller observed mean estimates, averaged across 172 meta-analyses and 63,730 estimated PCCs, are consistent with less overall bias.

3. ALTERNATIVE META-ANALYSIS ESTIMATORS OF PCCS

We investigate the four most promising adjustments for these small-sample biases of the meta-analysis of PCCs: the RE and the UWLS mean estimates of PCCs transformed to Fisher's *z*, *REz & UWLSz,* Hunter and Schmidt's[13] approach to the meta-analysis of correlations, *HS*, and lastly an adjustment to the degrees of freedom using UWLS, $UWLS_{+3}$.

3.1 Fisher's *z* Transformations: *REz & UWLSz*[ii]

The usual approach to avoid a mechanical correlation between estimated correlations and their variances is to first use the Fisher's *z* transform.[4] As Fisher[39] noted a century ago, what is true for correlations is true for partial correlations after degrees of freedom are appropriately adjusted. Fisher's *z*s are approximately normally distributed and their variance, $1/(df_i - 1)$, does not depend on the specific value of Fisher's *z*. Thus, the cause of meta-analysis' small-sample bias is avoided as the variances of Fisher's *z*s are independent of the specific values of Fisher's *z*. In brief, the standard solution is to convert all PCC's to Fisher's *z*, $z = 0.5 \cdot \ln[(1 + r_p)/(1 - r_p)]$, calculate the random-effects (or UWLS) estimate of the mean, $\bar{z}$, and lastly convert this back to PCCs, $REz = \frac{exp(2\bar{z})-1}{exp(2\bar{z})+1}$.

3.2 The Hunter-Schmidt Approach to the Meta-Analysis of Correlations

Hunter and Schmidt[13] offered an alternative meta-analysis approach (HS) to bivariate correlations, which they argued to be superior to RE*z*. HS uses the sample size, n_i, of each study as the weights. Thus, like Fisher's *z*, HS avoids any dependence arising from the weights being dependent on the estimated effect sizes and thereby their sampling errors. The HS meta-analysis estimate of the mean partial correlation is:

$$\overline{r_p} = \Big/_{\sum(n_i)}^{\sum\left(n_i r_{p_i}\right)} \qquad \text{i= 1, 2, …, } k. \tag{5}$$

The SE of HS is not calculated in the conventional manner by the square root of the inverse of the sum of the weights, as RE and FE do, but rather as:

$$SE_{\overline{r_p}} = {SD}/{\sqrt{k}} \tag{6}$$

Where PCC's standard deviation, SD, is computed by the square root of the weighted sum of squared deviations from the mean, $\overline{r_p}$:

$$SD^2 = {\sum\left(n_i(r_{p_i} - \overline{r_p}\right)^2)}/{\sum(n_i)} \qquad \text{i= 1, 2, …, } k \tag{7}$$

[13,40].[iii] Because HS uses sample size rather than inverse variance weights, it further addresses the concern that inverse-variance weights can be distorted when reported standard errors reflect modeling choices rather than pure sampling uncertainty.[41] Because the variance of Fisher's z is $1/(df_i - 1)$, UWLSz weights each estimate by approximately the sample size, as HS does; the two differ mainly in whether averaging is done on the z or the r scale.

3.3 Adjustment to Degrees of Freedom: $UWLS_{+3}$

Recall by Eq. (1) that PCCs are calculated from the estimated regression coefficients' *t*-values and degrees of freedom, *df*. Because degrees of freedom are so central to the use of PCC in meta-analyses, perhaps a simple adjustment to the degrees of freedom will correct the small-sample bias of conventional meta-analyses. Simulations confirm that by merely adding 3 to *df* in Eq. (1) the small-sample meta-analysis bias is reduced to scientific negligibility.[11] We call the resulting weighted average '$UWLS_{+3}$.'

To be clear, $UWLS_{+3}$ substitutes degrees of freedom that are three larger than the multiple regression's degrees of freedom into PCC's transformation formula, Eq. (1), and uses S_1^2, Eq. (2), as the variance. That is, $UWLS_{+3}$ first calculates each PCC as:

$$r_p^{+3} = {t}/{\sqrt{t^2 + df_{+3}}} \tag{8}$$

for $df_{+3} = n - s + 2$ with *s* as the number of independent variables in the multiple regression. Then, the meta-regression of r_p^{+3}/S_1 on $1/S_1$ is run, Eq. (4), to obtain the $UWLS_{+3}$ estimate. In their simulations, Stanley, Doucouliagos and Havranek[11] found that $UWLS_{+3}$ has smaller bias than REz and other conventional meta-analyses of PCCs.

3.4 Upward Bias Hypothesis: Comparing Alternative PCC Estimators

The known biases in the meta-analysis of partial correlation coefficients operate asymmetrically. That is, both small-sample bias and PSB systematically inflate meta-analysis estimated mean effect sizes. PSB is "one of the strongest findings across the sciences",[42(p370)] and dozens of surveys have found evidence of publication selection bias widely throughout the sciences, especially economics.[23,31] No analogous mechanism is known that would cause a typical meta-analytic estimator to *underestimate* the mean PCC systematically. In the absence of countervailing downward pressures, the direction of net bias is therefore one-sided: all estimators are plausibly too large, and the estimator that is, on balance, smallest is the one most consistent with the least upward distortion (*i.e.,* the *upward bias hypothesis).*

To clarify, bias is a theoretical property, equal to the difference between the expected value of an estimator's distribution and the value of the population parameter being estimated. Bias cannot be inferred from a single observed estimate nor does the fact that one estimator is smaller than another in a single application imply that it has smaller bias. To do so would be to confuse potential random sampling error or the effects of a singular realization of random heterogeneity as established fact. However, when an estimator is applied across a large and diverse set of independent meta-analyses (172), differences between its grand average and another estimator's are the empirical counterparts of differences in their expected values. Such differences need not reflect only differences in bias. Under random heterogeneity, simulations have shown that alternative weighting and different scales (z or r) produce different expected values; however, previous simulations have also revealed the biases of these alternative estimators with or without publication selection bias.[10-12] Smaller average estimates therefore show that an estimator is more conservative; they are consistent with, but do not establish, less upward bias. The upward bias hypothesis gains credibility only insofar as the observed ordering of estimators matches the ordering found in simulations in which the true effect is known.

We make no claim that one can *infer* smaller bias by observing smaller average estimators. To do so would be to commit the fallacy of 'affirming the consequent.' One can never make an inference from empirical evidence to universal or theoretical 'truth.' Hume argued that this is impossible when discussing the problem of induction.[43,44] What empirical evidence can do is to

provide corroborating support for a hypothesis when the direction of the evidence aligns with its predictions and when no plausible competing explanation accounts for the pattern as well. Furthermore, as we will see below, the empirical patterns seen in our survey are consistent with simulation studies with known small-sample and PSB biases.

4. AN ILLUSTRATION

Stanley *et al*.[7] conducted a meta-analysis of the partial correlations of the effect of information and communication technology (ICT) on economic growth. Although there is a strong consensus that the adoption and spread of these technologies are beneficial to the economy, some economists have questioned their practical effect. "You can see the computer age everywhere but in the productivity statistics".[45(p36)] Also, there is no universal measure of economic growth or of ICT. ICT consists of cell phones, broadband or other internet services, computer hard- and software, and now AI, while the most common economic outcomes are: GDP per capita, GDP growth rates, and increases in productivity. Economists who use productivity believe that increases in productivity will be reflected in economic growth, perhaps with a lag. Thus, with several different scales and ways to measure ICT and its economic impact, partial correlations are needed to measure the overall economic effect of ICT.

Using the data in Stanley et al.,[7] conventional random effects estimate the mean partial correlation between increases in ICT and economic growth as: 0.246, 95% CI (0.227, 0.265), and k = 416. Using REz does not notably affect these values: 0.251: 95% CI (0.232, 0.272).[10,11] According to conventional guidelines, both estimates are interpreted as 'small' effects ($.1 \leq r_p \leq .3$).[46] UWLS's estimate of the mean effect is 20% smaller, 0.195; 95% CI (0.177, 0.212) but remains what is widely considered a small effect. $UWLS_{+3}$ reduces UWLS by a trivial amount, 0.194; 95% CI (0.176, 0.211) but is 0.052 smaller than random effects. Lastly, both UWLSz and the Hunter and Schmidt (HS) approach find yet smaller estimates, 0.179; 95% CI (0.162, 0.196) and 0.172; 95% CI (0.157, 0.187), respectively, which is 27-30% smaller than RE. If instead we were to use the variance and SE formula known to cause larger biases, S_2 from Eq. (3), the results are larger than those from S_1: RE=0.253; 95% CI (0.233, 0.273) and UWLS=0.242; 95% CI (0.219, 0.264). Also note that these ICT estimates line up in the order seen in simulations where exact biases are known (the exception is RE and REz, whose order is reversed here).[11,12] Although

all these estimates are typically considered to be small, a 20-30% reduction in estimated effect size can have notable consequences in application for some moderators and meta-analyses, and it could, in some areas, alter policy recommendations and actions.

Although decreases of small-sample biases may be responsible for some of these reductions (recall the very small difference between UWLS and $UWLS_{+3}$ and between RE and REz), reductions in publication selection bias can explain why UWLS and HS estimates are noticeably smaller than RE and REz, a pattern also seen in simulations. That is, the pattern and differences among these estimators' average magnitudes are consistent with the notion that PSB is the more severe bias in this area of research. This interpretation is also consistent with tests for PSB. Both the Egger test and the proportion of statistical significance test (Egger: $z= 5.97$; $p<.0001$; PSST: $z= 6.73$; $p<.0001$) show clear evidence of PSB. The proportion of statistical significance test (PSST) is a test for the presence of PSB, and it cannot be interpreted as a small-study effect.[47] In general, UWLS has been shown to be less biased than RE when there is publication selection bias because its inverse variance weights are not moderated by RE's estimate of the heterogeneity variance.[24,25,27-32] As simulations have established, HS is expected to have smaller publication selection bias, because its weights are sample sizes, and PSB tends to be larger for small-sample studies.[12]

Of these alternative estimates of the mean partial correlation of ICT and growth, UWLSz and HS are the smaller, a pattern consistent with past simulations when there are both PSB and small-sample biases exaggerating these MA estimators. Nonetheless, it is important to recognize that none of these alternative methods produce differences sufficiently large to change how researchers interpret the overall, unconditional effect of ICT on economic growth.[iv] In all cases, we find that ICT has a small positive average effect on economic growth regardless of which meta-analysis estimator is used.

5. DATA

Our data were extracted from a large meta-research survey of statistical power and publication selection bias published in top-rated economics journals, as well as other scholarly journals, books, and unpublished working papers.[15] An extensive literature search for meta-analyses of economics research was conducted across: Econlit, Scopus, and Google Scholar. When the data were not

publicly available (109 meta-analyses), 74% of the contacted authors shared their data.[15] This process resulted in 368 meta-analysis datasets with 167,753 estimated effect sizes of all types. From these meta-analyses, we identified 172 meta-analyses that synthesized partial correlations as their effect sizes. Together, these 172 meta-analyses contain 63,730 estimated PCCs and their SEs extracted from 6,191 papers (or approximately 371 estimated PCCs per meta-analysis with 10 per study). Because our objective is to compare alternative meta-analysis estimators across 172 independent research literatures, we apply each estimator without modeling statistical dependence among estimates within studies. Within-study dependence affects the standard errors and confidence intervals of the estimates for a given meta-analysis. Independence across the 172 different meta-analyses allows our paired comparisons to account for the correlation between the estimates that different estimators produce from the same meta-analysis data. Nonetheless, studies with multiple estimates, whether independent or dependent, have more influence on all of these alternative estimators than they would with a single estimate. If these multiple estimate studies are also highly precise, they will influence HS and the UWLS family to a greater degree than RE or the simple mean. We report the mean proportions of total weight attributable to the highest-weight study (*i.e.,* research paper) and the three highest-weight studies across these alternative estimators in Appendix Table 1.

However, it is the sample sizes of the primary studies that drive the small-sample biases documented in previous papers and that is the central object of this study. The median of the median sample sizes is 210 while the median of the mean sample sizes is over twice as large, 571, reflecting highly skewed distributions of sample sizes. This is further seen in the average interquartile range = 3,789-316= 3,473. Large sample sizes and wide dispersions among sample sizes both lessen small-sample bias.

Next, we investigate whether alternative approaches to the meta-analysis of PCCs make noticeable differences in application. Although these meta-analyses come from economics, the statistical properties of PCC estimators depend on sample sizes, heterogeneity and selective reporting rather than on discipline. Thus, the small-sample biases and relative performance of estimators documented here will likely apply to other fields where PCCs have similar sample size distributions and heterogeneity.

6. RESULTS AND DISCUSSION

Table 1 reports the mean, median, standard deviation, frequency smallest, and mean squared difference (MSD-PP) from PET-PEESE (the precision-effect test and precision-effect estimate with standard error) for these alternative meta-analysis estimates of the mean PCC across these 172 meta-analyses.[v] For ease of comparison, the median PCC in each area of research is forced to be positive. That is, for inverse correlations identified by having a majority of the PCCs < 0, we multiply all PCCs in that meta-analysis by -1. As a result, when present, both small-sample bias and PSB will be positive, exaggerating the magnitude of reported estimates and thereby the weighted averages of these selectively reported estimates. Thus, the smaller average estimates are consistent with smaller biases through the upward bias hypothesis.

Table 1: Descriptive summary statistics

	Averages									Corrected
Statistic	Mean	RE1	RE2	UWLS1	UWLS2	$UWLS_{+3}$	HS	REz	UWLSz	PP
Mean	0.131	0.126	0.129	0.110	0.140	0.109	0.096	0.130	0.101	0.090
Median	0.097	0.087	0.090	0.061	0.069	0.061	0.056	0.086	0.057	0.040
Std Dev	0.115	0.116	0.117	0.128	0.175	0.127	0.109	0.122	0.117	0.171
Smallest	22	8	1	0	9	1	97	5	29	–
MSD-PP	0.022	0.020	0.021	0.010	0.019	0.009	0.010	0.021	0.009	–
Flipped	0.078	0.073	0.075	0.064	0.090	0.063	0.055	0.075	0.058	0.046

Note: Mean is the simple unweighted average. RE, UWLS, and HS denote random effects, unrestricted weighted least squares, and Hunter and Schmidt approach, respectively. RE and UWLS are numbered by the standard error used; 1 and 2 denote the standard error from Eqs. 2 and 3, respectively. 'z' refers to Fisher's *z* transformations. REz and UWLSz exclude 105 estimates (in two meta-analyses) with df ≤ 1. PP is the PET-PEESE estimate (PEESE if the PET intercept is positive and statistically significant at the 10% level, one-tailed, and PET otherwise; both use the standard errors of Eq. 2). 'Smallest' displays the frequency that a given estimator is the smallest weighted average, and MSD-PP is the average squared difference between each weighted average and PP. Flipped are the corresponding means for the subset of 50 MAs where the majority of the estimates were negative. 'Corrected' denotes correction for publication bias.

Consistent with past simulations, RE and UWLS are larger when S_2, Eq. 3, is used rather than those calculated using S_1, Eq. 2. Although the average difference of RE1 and RE2 is quite small, it is nonetheless statistically significant (paired $t = 8.76$; p<.0001). Also, as expected, the RE estimators are larger than the UWLS family of estimators: UWLS1, $UWLS_{+3}$, and UWLSz but

not as large as the simple unweighted average, the 'Mean'. With PSB, the magnitude of RE has been generally shown to be between UWLS and the unadjusted mean, and RE moves closer to the mean with higher heterogeneity.[27-29]

The differences between the UWLS and RE families are due to the higher weights that UWLS and HS assign to the more precise studies (Appendix Table 1). Under UWLS, UWLSz and HS, the highest-weight study (*i.e.*, a research paper) carries about one-third of the total weight, roughly twice the share that RE or the unweighted mean receive, and the three highest-weight studies carry not quite twice the weight for the UWLS family. Yet even the unweighted simple mean gives the highest-weight study 16% of the weight because some papers report many estimates. For this data, the most precise studies tend to have smaller PCCs among these 172 meta-analyses as evidenced by the positive and statistically significant Egger regression coefficient on SE (see below) and as consistently seen in simulations with PSB. When the most precise studies tend to have smaller effects, UWLS will be smaller than the corresponding RE or the simple unweighted mean estimate. Further 'overweighting' of the most precise estimates can be helpful when there is a serious risk of publication selection bias. The advantage of this 'overweighting' is vividly seen by the weighted average of the adequately powered (WAAP), which calculates UWLS across only those estimates that have 80% or higher retrospective power.[29] WAAP, thereby, gives greater weight to the most precise studies than any of the methods reported here. Yet, WAAP has the smallest bias rank compared to two dozen alternative approaches to PSB and across 1,665 benchmark simulations originally proposed by four independent teams of researchers. These benchmark simulations are claimed to provide a neutral framework for evaluating alternative PSB correction methods.[32]

The choice of PCC variance makes a larger difference to the estimated mean effect as calculated by UWLS (0.03) because a few large and highly precise estimated PCCs can carry considerable weight when using S_2.[vi] As shown by Stanley and Doucouliagos[10] and corroborated here, there is little reason for the continued use of the conventional PCC standard error formula, S_2, in the meta-analysis of PCCs or correlations.[12]

Without S_2, the means range from 0.096 to 0.130, while the median varies from 0.056 to 0.087. Note that all medians are smaller than 0.1 and the reduction of the medians relative to their corresponding means is greater than the differences seen across these alternative approaches to the

meta-analysis of PCCs. Similar to the ICT illustration, above, RE's and REz's mean and median are both larger than the respective values of HS and the family of UWLS estimators. Again, HS is the smallest, on average, and it is the smallest most frequently (Table 1). Interestingly, the highest correlation (0.987) among these methods is between HS and UWLSz. Although Fisher's z version of random effects has often been suggested as the preferred method,[49] it has the largest mean, slightly greater than RE. Therefore, random-effects using Fisher's z is the least conservative and thereby consistent with being the most biased estimate of the mean among these alternative approaches that use S_1.

One may prudently question whether smaller is truly less biased. Of course, random sampling errors and/or the random influence of one or a few extreme estimates can cause some of these individual estimates across 172 meta-analyses to be smaller (or larger) than the true population mean effect. However, any notable systematic underestimation of the 'true' mean effect by these weighted averages is rather unlikely where there is no known downward bias.[11,12] Furthermore, there are clear indications of publication selection bias (PSB) in this survey that, on average, exaggerate the reported magnitudes of mean effects. Nonetheless, we cannot logically infer less bias from smaller average estimates as that would commit the error of 'affirming the consequent.' Rather, these smaller average estimates exhibit a pattern consistent with the 'upward bias hypothesis' and with past simulations where we know that smaller is, in fact, less biased.

To confirm the likely presence of PSB in our survey data, we first ran an aggregate fixed-effect-panel Egger regression with cluster-robust SEs across 172 MAs. The meta-regression coefficient on SE, S1 from Eq. (2), is 0.850; 95% CI = (0.442, 1.26). This panel meta-regression model allows every area of research to have a different mean effect; however, the PSB effect (estimated to be 0.850 standard errors) is assumed constant across MAs. Although rather small,[26] the PSB effect is clearly positive and statistically significant ($t = 4.08$; $p < 0.001$). In contrast, the proportion of statistical significance test (PSST) allows both the mean effect and the intensity of PSB to vary freely across any number of MAs and presumes no specific mechanism of selection or preference for statistical significance and, thereby, cannot be interpreted as a small-study effect.[47] Also, PSST and its constituent elements (excess statistical significance, ESS, and the expected number of statistically significant reported effects in the absence of any PSB, Esig) are easily, meaningfully, and naturally aggregated across meta-analyses in meta-research

analysis.[14,15,47] PSST is generally a more powerful test of PSB as we see here, where PSST z = 23.2 ($p << 0.001$). Thus, there is clear evidence for selection for statistical significance in some studies among these 172 meta-analyses. When we look at 172 individual Egger tests for PSB, 47.1% are statistically significant in a positive direction ($\alpha = 0.05$) while 39.5% of PSSTs are statistically significant at the same level (both tests one-sided). Thus, this survey contains clear evidence that is consistent with notable selection for statistical significance of the type that leads to upward biases. Note that the smallest average estimates (HS and the UWLS family of estimators) are also the closest (i.e., smallest MSD) to the PET-PEESE corrected estimates across these 172 meta-analyses (Table 1). Because PET-PEESE is merely an estimator, the closeness of HS and the UWLS to these corrected estimates does not prove that they are less biased.[vii] Nonetheless, this is further evidence consistent with the upward bias hypothesis.

The meta-research findings of these PCC meta-analysis methods, as reported in Table 1 and seen in practice, are fully consistent with relative magnitudes of these estimators seen in previous simulations, including the expectation that HS will be both the smallest and the least biased.[10-12] Although not all of these estimators were simulated, those that were have the same ordering from smallest to largest when there is known publication selection bias. For the smallest population mean (0.11) simulated, the average biases across the 50% and no PSB heterogeneity conditions are: 0.006 for PP, 0.031 (HS), 0.038 ($UWLS_{+3}$), and 0.042 for REz.[12] The relative magnitudes found among our survey's average estimates relative to each other and to PET-PEESE (PP) match what was seen in simulations. However, as expected, the precise magnitudes of these differences cannot be expected to be the same because the observed conditions do not exactly match the simulations. For example, from the aggregate estimates of ESS (excess statistical significance), we estimate that the incidence of PSB is $Psss = \frac{ESS}{1-Esig} = \frac{0.04455}{1-0.3743} = .071$.[15] Though present, this is a lower estimate of the incidence of publication bias than the average of the 50%-PSB and no-PSB conditions imposed upon the simulations. Thus, we would expect to see smaller apparent PSB biases in this survey than found in past simulations. Because 0.009 is the largest reduction in UWLS seen among either RE's or UWLS's small-sample bias corrections, most of the small differences seen across these alternative MA methods are therefore more likely to reflect PSB than small-sample bias.

The strong performance of HS is also consistent with growing evidence that reported precision in observational research can be endogenous to researchers' modeling choices.[41] Recent research focused on the small-sample biases of PCC meta-analyses and showed that they could be driven to negligibility through alternative approaches, REz and $UWLS_{+3}$. Here, we find that these corrections for PCC's make no noticeable reduction to these small-sample biases—compare UWLS to $UWLS_{+3}$, or UWLSz, and RE to REz. Thus, the small differences among methods are likely to reflect a reduction of PSB, and they are also consistent with other simulation studies that compared the performances of RE and UWLS.[24,25,27-32] Also, see the "living synthetic benchmarks" simulation comparisons compiled from over 1,600 previously published simulation designs.[32]

When we focus only on those MAs where the majority of the PCCs < 0 (Flipped, Table 1), all averages are consistently 0.04 to 0.055 smaller. This should not be surprising as these MAs are more likely to be drawn from populations with small or virtually zero mean effects. Nonetheless, we again find the same ordering from smallest to largest among these MA estimators that use S_1 as seen before (Table 1) and in simulations where there is known publication selection bias. The preservation and stability of this complex ordering of estimators' averages is further consistent with the explanation that the observed differences seen in this survey are due to their known differential responses to PSB and thereby consistent with the upward bias hypothesis.

Although often small, the differences among these meta-analysis methods are systematic and statistically significant— REz v RE (0.004; paired $t = 4.98$; $p < .0001$), RE v UWLS (0.016; $t = 3.56$; $p < .001$), UWLS v $UWLS_{+3}$ (0.001; $t = 6.67$; $p < .0001$), UWLS v UWLSz (0.009; $t = 5.96$; $p < .0001$), RE v HS (0.030; $t = 7.09$; $p < .0001$), UWLSz v HS (0.005; $t = 3.48$; $p < .001$), and $UWLS_{+3}$ v HS (0.013; $t = 5.48$; $p < .0001$). However, are these small numerical differences in correlations practically or scientifically notable?

The smallest effect size of interest is context specific. Nonetheless, a Cohen's d of 0.2 is commonly used when lacking specific theoretical or policy guidance to the contrary. Because we are evaluating 172 very different areas of research without a common cost-benefit profile, we have no choice but to default to widespread practice, which is Cohen's guidelines for a 'small' (or nontrivial) correlation $\geq$ 0.1 (equivalent to Cohen's $d \geq 0.2$).[46] Partial correlations less than 0.1 contribute less than 1%, marginally, to the explanation of the phenomenon in question. With a few

exceptions where there are large-scale and low-cost applications, PCCs smaller than 0.1 would be widely regarded as trivial. In a field as noisy as observational economics research, accounting for less than 1% of the relevant variation is trivial. By this guideline, the majority of economics research investigates empirically trivial phenomena, at least as represented by these 172 meta-analyses. Relative to RE, 55% of these areas of research are scientifically trivial, 66% for UWLS and $UWLS_{+3}$, and 67% for UWLSz and HS.

How often do these corrections make a notable difference in how we might interpret the average effect in an area of economics research? That is, how often would a nontrivial average effect as calculated by RE, the conventional approach, become scientifically trivial when these small-sample bias reductions are applied? Between 32 and 33% of small RE estimates ($0.1 < r_p < 0.3$) are reduced to triviality when replaced by UWLS, $UWLS_{+3}$, UWLSz, or HS, while the reverse happens once, < 2%. Thus, UWLS and HS can make a notable difference at the margins in practice.

An important limitation is that all of these alternative MA estimators are estimates of the *unconditional* mean effect. "(A)ny simple overall meta-analysis needs to be interpreted with caution if there is publication selection bias and/or heterogeneity in the reported estimates".[7(p713)] Nonetheless, high heterogeneity is typical of economics and the social sciences. According to our PCC survey, the average I^2 is 86% and $\hat{\tau}^2$ is 0.026. All but one MA has statistically significant heterogeneity. With heterogeneity this high, one might question whether any single mean estimate can be representative in the majority of these areas of research. Because notable heterogeneity is widely recognized and generally assumed by economists, almost all economic meta-analyses code dozens of research dimensions to accommodate heterogeneity through multiple meta-regression analysis (MRA), thereby providing conditional estimates of mean effect. Although any simple overall average may not be relevant to many economic meta-analyses, they remain relevant to medicine and psychology, where mainly unconditional meta-analysis averages are reported. Even for economics, biases found among the unconditional mean estimates are likely to propagate into the MRA, thereby distorting the estimated effects of moderators. The central point to this survey is to understand whether the small-sample biases uncovered in recent studies are likely to affect the interpretation of meta-analysis findings in practice, and we find that they generally do not.

Another potential caveat is that the typical PCC in these 172 meta-analyses is quite small, median simple average PCC = 0.097, and small-sample biases are larger with larger PCCs.[10,11] However, even for the larger PCC meta-analyses, we find that the small-sample bias is negligible. Only one meta-analysis has a difference between UWLS and $UWLS_{+3}$ > 0.01 (0.011). Nonetheless, it would be prudent for meta-analyses of large PCCs to routinely use one of these corrections for small-sample bias as a robustness check.

UWLS, UWLS+3, and UWLSz can be directly incorporated into MRA without any further alterations. However, there is no known MRA extension for HS. Developing such an extension is a promising direction for future research. Thus, UWLS is likely the most practical PCC approach for economics applications. As a simple rule of thumb, we recommend that meta-analysts of PCCs report UWLS and, when only an unconditional mean is needed, also HS or UWLSz.

7. CONCLUSION

This study surveys the findings from alternative meta-analysis methods across 172 meta-analyses of partial correlation coefficients, PCCs. The primary purpose of this survey is to assess whether recently identified biases of PCC meta-analyses and the associated methods offered to reduce these biases would likely make a noticeable difference to the practice of meta-analysis.[9-11]

Van Aert and Goos[9] argued that meta-analyses using standard errors calculated from Eq. (2), S_1, would be biased relative to using the 'correct' standard errors, S_2, Eq. (3). Stanley and Doucouliagos[10] demonstrated that the exact opposite was true; that is, conventional meta-analysis methods (RE, FE, and UWLS) using S_1 have consistently smaller small-sample biases than those calculated from S_2. This survey supports the advantage of using S_1 rather than S_2 as the PCC's SE because the averages for both RE and UWLS across 172 PCC meta-analyses increase when S_2 is used in place of S_1. Recall that the way these meta-analyses are aligned (inverse associations made positive) implies that both small-sample bias and publication selection bias (PSB) if present will exaggerate the magnitude of reported effects. The estimators with smaller average effect sizes (those using S_1) are *known to be less biased in simulations* and thereby are consistent with smaller biases (i.e., the upward bias hypothesis) especially as there is ample evidence of PSB in our survey

data. Because these empirical findings are consistent with theory and simulations, we see little reason to continue using the conventional standard error, S_2, in the meta-analyses of PCCs.

To correct the small-sample biases identified in recent studies, Stanley, Doucouliagos and Havranek[11] suggested a simple correction for degrees of freedom, $UWLS_{+3}$, and the well-known Fisher's z because its variance does not depend on the PCC estimate. However, these corrections, REz, UWLSz, and $UWLS_{+3}$, do not notably differ from their corresponding unadjusted meta-analyses, RE and UWLS, and thereby make little scientific or practical difference to how PCC meta-analyses would be interpreted or understood in practice. As discussed above, the sample sizes and complexity of these 172 PCC meta-analyses wash out any notable small-sample bias. What little remains is likely publication selection bias (PSB). This is consistent with the fact that UWLS is smaller and less biased than the corresponding random effects estimator in simulations when there is known PSB. In particular, UWLS's median is 30% smaller than RE's, consistent with the relative performance of these estimators found in numerous simulation studies when there is PSB.[24,25,27-30,32] Also, notable PSB is confirmed by the proportion of statistical significance test (PSST)[47] and the Egger meta-regression.

Stanley, Doucouliagos and Havranek[12] also investigated the Hunter and Schmidt (HS) approach to the meta-analysis of bivariate correlations. In this survey of PCC meta-analyses, HS is the smallest and thereby consistent with the upward bias hypothesis' association of smaller with less biased and with Hunter and Schmidt's[13] argument that their meta-analysis approach is preferable to RE using Fisher's z. Although HS is typically a little smaller than the UWLS family of estimators, this difference will rarely be meaningful in practice. Besides, the medians of HS and UWLSz are practically the same. As shown in previous simulations, meta-research, and consistent with this survey, HS and the UWLS family tend to be less affected by publication selection bias than RE.[10-12,24,25,27-30,32,50] However, extreme estimates can differentially affect these methods in specific areas of research. When in doubt, report the more conservative (i.e., smaller and less statistically significant) estimate and emphasize whichever estimator has the best demonstrated performance under likely biases.

The only meaningful difference among any of these estimators is at the margins when compared to RE. When 0.1 is regarded as the smallest effect size of interest, as discussed above,

approximately one-third of the 'small' average effects identified by RE become 'negligible' by UWLS, $UWLS_{+3,}$ UWLSz, and HS.

In summary, this survey supports the use of S_1 over S_2 and UWLS and HS over RE in the meta-analysis of partial correlation coefficients. Otherwise, these corrections for the small-sample bias of the meta-analysis of partial correlation coefficients matter little in practice, at least as seen in our large sample of economic meta-analyses. However, these differences may still be of consequence for some areas of experimental research where the sample sizes are routinely small. We encourage researchers in health sciences, psychology, and environmental studies who meta-analyze PCCs to compare these estimators using data from their own fields.

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None

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The data and code used in this survey are available at: https://github.com/PetrCala/pcc-survey, with a summary of the results at https://meta-analysis.cz/pcc_survey/

Artificial intelligence use: Claude Code with Claude Sonnet 5.5 (Anthropic) and GPT Codex (OpenAI), assisted in checking and replicating the computations, cross-checking the results, and editing the text of this revision. The authors are responsible for all of the paper's content.

Appendix:

Appendix Table 1: Further descriptive summary statistics

	Averages								
Statistic	Mean	RE1	RE2	UWLS1	UWLS2	$UWLS_{+3}$	HS	REz	UWLSz
Mean	0.131	0.126	0.129	0.110	0.140	0.109	0.096	0.130	0.101
Median	0.097	0.087	0.090	0.061	0.069	0.061	0.056	0.086	0.057
Std Dev	0.115	0.116	0.117	0.128	0.175	0.127	0.109	0.122	0.117
W_top 1	0.164	0.172	0.170	0.339	0.345	0.339	0.333	0.173	0.343
W_top 3	0.352	0.366	0.364	0.590	0.595	0.590	0.584	0.368	0.593

Note: Mean is the simple unweighted average. RE, UWLS, and HS denote random effects, unrestricted weighted least squares, and Hunter and Schmidt approach, respectively. RE and UWLS are numbered by the standard error used; 1 and 2 denote the standard error from Eqs. 2 and 3, respectively. 'z' refers to Fisher's z transformations. W_top 1 is the share of the total weight given to the highest-weight primary study, and W_top 3 is the share given to the three highest-weight primary studies, both averaged across the 172 meta-analyses. A study's weight is the sum of the weights of its estimates.

[i] 'Fixed effect' in a meta-analysis context has an entirely different meaning than 'fixed effects' in panel data models. In panel meta-regression models, fixed effects are the general case while random effects are applicable only when study effects are confined to be normally distributed and the meta-regression coefficients are constant across studies.[5,22] Fixed-effects meta-meta-models can offer stronger quasi-experimental designs for meta-research.[14]

[ii] We thank an anonymous reviewer for suggesting that we report the UWLS version of using Fisher's z transformation. In practice, it gives estimates very close to HS—see below.

[iii] Replacing sample sizes, n_i, with PCCs' degrees of freedom is also sensible. However, we also suspect that such nuances will not matter in practical applications which have much larger biases (PSB) and inefficiencies. In the 172 meta-analyses that we investigate, we use degrees of freedom because not all authors reported sample size. Fortunately, degrees of freedom can always be calculated from Eq. (1) when we know r_p and its SE, both of which are required in meta-analyses.

[iv] It is not clear that the unconditional mean in this application carries much inferential value as heterogeneity is quite high. Because these estimators are averaged across PCCs employing very different measures of economic growth (per capita GDP, GDP growth rates, and productivity), it is unclear whether any mean is a meaningful representation of this area of research. The meta-analysis from which this illustration is taken agrees. "(A)ny simple overall meta-analysis needs to be interpreted with caution if there is publication selection bias and/or heterogeneity in the reported estimates".[7(p713)] Almost all economic meta-analyses of PCCs, including this one, are primarily meta-regression analyses (MRA) that explicitly control for observed differences of measures and methods, among other factors, and rely on the MRA, conditional, findings for their conclusions. Here, we are merely illustrating the typical pattern found in simulations among these alternative MA estimators and across 172 MAs of PCCs, as discussed below. We make no inference about the 'true' unconditional average PCC of ICT.

[v] We thank an anonymous reviewer for suggesting that we calculate the 'smallest' and the closeness to PET-PEESE.

[vi] When one or few estimates notably change RE or UWLS, they are potentially influential (*i.e.,* leverage) points and should be further investigated. Experience suggests that the specific estimate that causes notable changes to the magnitude of the average effect is often a coding error, typo, or a conceptually different but related estimate. Because the purpose of this empirical study is to survey how different MA methods affect PCC meta-analyses in practice, we did not attempt to further remove outliers or influential points. In a survey of 358 behavioral science meta-analyses, alternative treatments of outliers barely moved the mean effect but sometimes reversed statistical significance, mainly for results near the threshold.[48]

[vii] Although PET-PEESE calculated using PCCs, as reported in Table 1, can overcorrect in individual meta-analyses, it does not do so, on average, in simulations.[12] PET-PEESE calculated from Fisher's z does have a small overcorrection in these simulations, but it also has better MSE and coverage, on average.